

APPENDIX F

CALIBRATION

A. Sampling Pump Calibration

Calibrate personal sampling pumps before and after each day of sampling using one of the techniques described below. Assure that the calibration equipment is within its prescribed calibration interval, and record the serial number of the calibration equipment in your case file and the OIS air sampling worksheet. The SLTC's chemists sometimes use sampling pump calibration data to verify air sample volumes.

If the sampling pump is equipped with a rotameter or digital flow readout, record the reading in the OIS air sampling worksheet. Bear in mind that the accuracy of a pump rotameter is only approximate; it is intended primarily to facilitate setting the flow rate for calibration.

Most of the following examples in this appendix use filter cassettes as the sampling media, but the examples are generally applicable to adsorbent tubes as well.

NOTE: Precision rotameters are no longer used by OSHA for calibration due to the potential for measurement error (e.g., tests with precision rotameters have indicated significant error due to pump pulsation). Inverted burets may still be useful, but their use is discouraged because they are no longer considered a primary calibration standard.

Before pre-calibration, replace or recharge sampling pump batteries as needed. Check the rechargeable Ni-Cad batteries in older pumps before use under load (i.e., turn pump on and check voltage at the charging jack with the sampling device in-line).

Place the same **type** of sampling media in-line during sampling pump calibration that will be used to sample in the field. Do not use the actual cassette and filter intended for sampling use to perform calibration.

B. Pump Calibration for use with Cyclone

The "Jarless Cyclone Calibration" procedure is the recommended method for calibrating a cyclone. A one-liter jar should no longer be used due to technical issues such as leakage of the jar lid.

The purpose of the procedure is to determine whether the sampling pump will be able to maintain the required flow rate as the drop in static pressure grows due to particulates loading up on the filter. The typical pressure drop across a clean 5- μ m filter is 2 inches of water pressure. The additional pressure drop from the cyclone is approximately 0.25 inches. As a filter loads up, the additional pressure drop may be as high as 20 inches of water pressure.

The procedure is demonstrated in the video "Jarless Cyclone Calibration Method" on the OSHA intranet.

Below is a summary of the basic steps of the procedure:

- Adjust and calibrate the pump to a flow rate of 1.7 L/min with a light load attached;
- Increase the load to 25–35 inches and check to be sure the flow rate is within +/- 5%;
- Remove the load and attach the cyclone to the sampling pump, then verify that the load is at between 2 and 5 inches of water pressure;

Detailed Step-By-Step Instructions

1. Connect the sampling train:

- Connect the sampling pump to a Tee fitting.
- Connect the Tee to a pressure gauge and the light load (i.e., a clean 5- μ m filter or an adjustable bonnet valve).
- Connect the load to the air outlet of the pump calibrator (bubble meter).
- Turn on the sampling pump.
- If a bonnet valve is used, adjust to create a load of 2–5 inches of water pressure. If a 5- μ m filter is used, verify that the pressure drop is 2–5 inches of water pressure. If not, this may indicate leakage around the filter.
- Set the pump flow rate to 1.7 L/min.
- Take at least three flow readings and record the average flow rate. Readings should be within +/- 2% of each other.

2. Increase the load:

- If a bonnet valve is used, adjust to increase the load to 25–35 inches of water pressure. If a 5- μ m filter was used for the light load, remove it and replace it with a set of six 0.8- μ m filters connected in series (i.e., sandwiched in a four-piece cassette with four backup pads). If the load is not at least 25 inches of water pressure, ensure the cassette is tightly compressed to prevent air from bypassing the filters.
- Turn on the pump, and then check and record the average flow rate. If the flow is not within +/- 5% under increased load, return the pump to the CTC for repair with a note explaining the problem.

3. Add the cyclone to the sampling train:

- Disconnect the pump calibrator and remove the load.
- Attach the cyclone with a clean 5- μ m filter in place of the load.
- Turn the pump on and confirm that the pressure drop is 2–5 inches of water pressure with the cyclone attached.

Refer to the cyclone leak test and cyclone cleaning procedures as described in [Appendix I](#).

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C. Electronic Flow Calibrators

The Gilian Gilibrator® I and II by Sensidyne® (see Figure F-1) are electronic bubble flow meters, used to calibrate sampling pumps, that provide instantaneous air-flow readings and cumulative averaging of multiple measurements. These calibrators measure the flow rate and display the results as volume per unit of time (e.g., mL/min) and can be used to calibrate most air sampling pumps. Different flow cells are used to accommodate different flow ranges. The middle-sized flow cell is typically used for personal sampling for particulates, while the largest cell is used for high volume area sampling and the smallest cell may be needed for certain low flow sorbent tube methods. The total range with the different flow cells is from 1 mL/min to 30 L/min. Gilibrators **should not** be left plugged into the charger for extended time periods because doing so will decrease the service life of the battery.

Another wet-cell calibrator available in some OSHA field offices is the miniBuck™ as shown in Figure F-2. Its operation is similar to the Gilibrator.

The Bios Defender™ shown in Figure F-3 is an electronic dry-piston flow meter used to calibrate sampling pumps that provides immediate and average readings. The device can be used to calibrate either pressure (labeled inlet) or vacuum (labeled outlet) flow sources. The vacuum port is used to calibrate sampling pumps, and the pressure port is used to calibrate the outlet of sampling pumps used to fill gas sampling bags. The Bios Defender has a lead-acid battery and can be left on charge for an indefinite time without damaging the battery. Different models of the instrument cover an optimum flow range of 5 mL/min to 30 L/min.

FIGURE F-1. GILIAN GILIBRATOR PUMP CALIBRATOR



FIGURE F-2. MINIBUCK PUMP CALIBRATOR



If using a Bios Defender to calibrate an MSA Escort® ELF pump, use an isolating flow restriction providing at least 5 inches of water column pressure between the pump and the flow meter. For example, use air sampling media (cassette or sorbent tube) with a Gemini variable orifice. Failure to use such an isolation technique may result in +/- 2% calibration inaccuracy.

The CTC recommends that the Bios Defender not be used in a very dusty environment because dust that flows through the calibrator piston area has the potential to scratch the glass and piston inside the calibrator. The CTC also recommends that neither the Gilibrator nor the Bios Defender flow calibrator be used in corrosive or otherwise contaminated environments.

Properly functioning and calibrated Gilibrators and Bios Defenders have an accuracy of approximately 99%. Use the appropriate Gilibrator flow cell or the Bios Defender model with the appropriate range of airflow for the pump airflow to be calibrated.

It is recommended that the flow rates obtained from these devices be reported to three significant figures. For example, a flow rate shown as 1.006 L/min should be reported as 1.01 L/min.

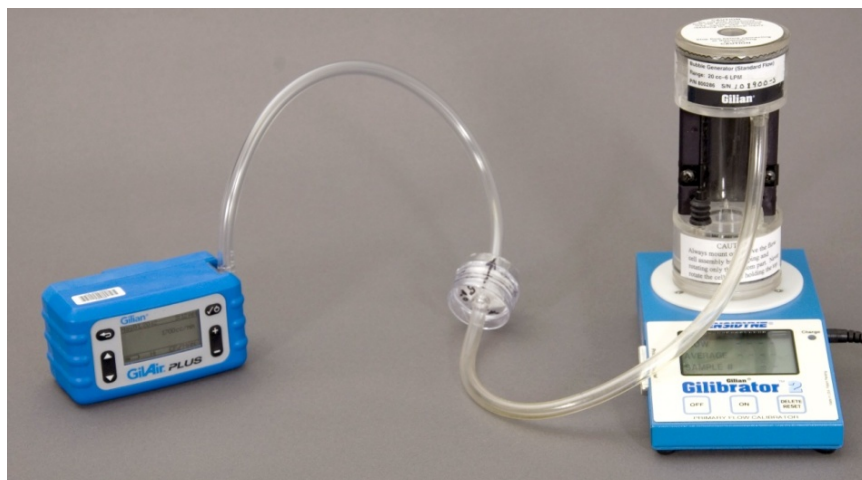
D. Procedures

NOTE: The following instructions and figures were written for the Gilian Gilibrator flow calibrator as shown in Figure F-4, however, the Bios Defender and miniBuck calibrator can be substituted in most cases (for use with MSA Escort ELF pumps, the isolating flow restrictor must be used as described above).

FIGURE F-3. BIOS DEFENDER PUMP CALIBRATOR



FIGURE F-4. CASSETTE ATTACHED TO ELECTRONIC BUBBLE METER FOR PUMP CALIBRATION



1. Perform the calibration at the pressure (altitude) and temperature where the sampling is to be conducted. If this is not possible, consult the operating manual for the sampling pump to determine if the air volume needs to be adjusted for temperature and pressure. Allow the pump to run for one to five minutes before voltage check and calibration. Consult the CTC regarding correction factors if sampling must be performed at freezing temperatures.
2. Connect the collection device, tubing, pump, and calibration apparatus as shown in Figure F-4 for the cassette (or sorbent tube) sampler. Note that cassette adapters (e.g., plastic or metal Luer taper adapters) should not be used. Luer adapters in front of a filter in a calibration train can potentially generate significant back pressure for which some pressure regulating pumps may not be able to compensate, resulting in inaccurate results. Luer adapters behind filters can affect sample distribution across an open-face cassette, and some Luer adapters are long enough that they may even make contact with the backup pad in the cassette. For this reason their use is not recommended. There are commercially available filter cassette holders with integrated connectors that do not have an adverse effect on back pressure.
3. Visually inspect all plastic tubing connections. Be certain that there are no leaks.
4. Gilibrator: Assemble the Gilibrator as per the equipment manual. Introduce soap solution into the flow cell through the air inlet boss. Add enough soap solution until the angled edge at the bottom of the bubble generator ring is immersed in the solution. After connecting the sampling pump and turning it on, push the button several times to wet the inside of the electronic flow cell with the soap solution.

Bios Defender: No preparation required.

5. Turn on the pump and adjust the pump rotameter and/or digital flow display (if so equipped) to the appropriate flow rate.
6. Gilibrator: Press the button on the electronic bubble meter. Visually capture a single bubble and electronically time the bubble. The accompanying printer will automatically record the calibration reading in L/min.

Bios Defender: Press and release the Read button for a single measurement. Press and hold the Read button for consecutive measurements.

7. Adjust the sampling pump to the correct flow rate using the calibrator results. Adjust the sampling pump while it is running.
8. Confirm that the sampling pump is adjusted properly and take additional calibrator readings. Three or more consecutive readings should be taken and should be within about 2% of each other and then averaged.

FIGURE F-5. BUBBLE ACTUATOR BUTTON ON GILIBRATOR



9. Repeat the procedures described above for all pumps to be used for sampling. The same cassette and filter may be used for calibrations involving the same sampling method. Do not use the actual cassette and filter intended for sampling use to perform calibration.

FIGURE F-6. GILIBRATOR DISPLAY PANEL SHOWING FLOW RATE, SIDE BY SIDE WITH GILAIR SAMPLING PUMP WITH DIGITAL FLOW READOUT



E. Instructions for Calibrating MSA Escort ELF Pumps with “Dry” Piston-Type Calibrators

When calibrating Escort ELF pumps with piston-type calibrators, extra steps must be taken to ensure that the pumps are calibrated accurately. The design of Escort ELF pumps make them susceptible to calibration inaccuracies due to pressure spikes created by the mechanical action of Bios International piston-type calibrators, such as the Defender series models. The rubber-walled pulsation dampeners incorporated into the Escort ELF pump design do not sufficiently compensate for the pressure spikes from the Bios calibrators; as a result, these pressure impulses can produce an uncertainty of over 4% ($\pm 4\%$) in the calibration accuracy of Escort ELF pumps. **THE ESCORT ELF PUMPS SHOULD NOT BE CALIBRATED USING THE BIOS INTERNATIONAL PISTON TYPE CALIBRATORS BECAUSE THEY CAN PRODUCE AN UNCERTAINTY ERROR OF OVER 4%** (compared to the desired margin of error of $\pm 1\%$). No other type of pumps are affected by this phenomenon, nor do the Escort ELF pumps suffer this susceptibility when wet cell bubble-type air flow calibrators are used or when the pumps are deployed with air sampling media or filter cassettes.

The manufacturers recommend that an isolating flow restrictor, such as a 0.030-inch orifice, be placed between the pump and the flow meter. The 0.030-inch orifice was shown at the CTC to create about 20 inches of water column (WC) of additional loading to the sampling pump at 2,000 mL/min. This additional loading causes the pump to speed up to maintain the required flow setting, which, in turn, makes the small changes in loading caused by the action of the calibrator's piston to have a negligible effect on the airflow within the Escort ELF pump. The use of a 0.030-inch orifice will ensure that there will be a sufficient load on the Escort ELF pump to mitigate the pressure spikes from the Bios piston-type calibrator across the entire effective flow-rate range of the pump, thus enabling the operator not to exceed the required 1 percent (± 1 percent) pump calibration accuracy.

F. Calibration Procedures for Open-Face Filters

Open-face cassettes are used for asbestos and certain chemicals such as isocyanates, crotonaldehyde, and glutaraldehyde.

1. The appropriate way to calibrate an open-face cassette is to use the cover section which comes with the cassette and attach the tubing directly from the electronic flow calibrator to the inlet port on the cassette cover. Be certain there are no leaks and do not use a Luer adapter. This set-up will provide the least amount of flow resistance and represent the open-face conditions while actually sampling.
2. Perform the pump calibration at the pressure (altitude) and temperature where sampling is to be conducted. If this is not possible, consult the operating manual for the sampling pump to determine if the air volume needs to be adjusted for temperature and pressure.

G. Calibration of Impingers and Bubblers

1. Set up the calibration apparatus as shown in Figure F-6, but instead of using a cassette, attach an impinger or bubbler filled with the amount of liquid absorbing solution specified in the sampling method. If a prefilter or cassette is described in the [CSI](#) file, include the correct filter in line. Include an adsorbing solution trap if one is to be used in air sampling.
2. Connect the tubing from the electronic bubble meter to the inlet of the impinger or bubbler.
3. Connect the outlet of the impinger or bubbler to the tubing to the pump. Be certain there are no leaks.
4. Calibrate the pump to the flow rate specified in the [CSI](#) file for the sampling method.

H. Maintenance and Care of Electronic Calibrators

Consult the manufacturer's instruction manuals for complete details. Periodically, compare the calibrator to another unit to make sure that it is functioning properly. Return the calibrator to the CTC annually to be calibrated and serviced.

Gilian Gilibrator:

1. For units that are used daily, connect the short length of storage tubing to the air outlet (upper) boss and the air inlet (lower) boss of the flow cell. This will prevent evaporation and concentration of the soap solution. Store in a clean area.
2. For units that will not be used for more than a week, remove the flow cell by unplugging the power/data cord and then gripping the bottom of the flow cell and rotating it one quarter turn counter-clockwise. Caution: gripping the top of the flow cell before turning it can stress and crack the flow cell. Then tip the cell horizontally and drain the soap solution out through the air inlet (lower) boss. Allow the flow cell to air dry completely before storage. Do not leave the calibrator plugged into the charger for extended time periods because doing so will decrease the service life of the battery. Store in the case to protect from breakage or dust accumulation.
3. Check the unit before use. Wipe the outside with a damp cloth if needed. If stored properly, routine cleaning is unnecessary. If there is excessive soap residue inside the flow cell, disconnect it and rinse with warm water. The acrylic flow cell can be easily scratched. Do not allow the flow cell (center tube, where sensors detect the soap bubble) to be scratched or to get dirty. Never clean the cell with acetone, alcohol or other cleaning solutions.

4. Leak testing. If leakage is suspected, perform a leak test as described in the Gilibrator manual. No leakage should be observed. Never pressurize the flow cell with more than 25 inches of water pressure. The CTC performs a leak check as part of their annual service.
5. Calibrator Calibration. The calibrator is factory calibrated using a standard traceable to the National Institute of Standards and Technology. The calibrator is linear throughout the entire range. The CTC will calibrate the unit as part of their annual service.
6. When transporting the calibrator, especially by air, it is important that one side of the storage tube which connects the air inlet and outlet be removed to equalize internal pressure within the calibrator. Do not transport the unit with soap solution or with storage tubing in place.
7. The calibrator soap is a concentrated and sterile solution formulated to provide a clean, frictionless soap film bubble over the wide, dynamic range of the calibrator. The sterile nature of the soap is important in order to prevent residue buildup in the flow cell center tube, which could cause inaccurate readings. The use of any other soap is not recommended. Proper soap solution is available from the CTC's expendable supplies program.

Bios Defender:

1. Do not use liquid solvents or abrasive cleaners to clean the calibrator; wipe only with a cloth lightly dampened with water. Store the instrument in a clean, dry place and with the unit on charge, if possible.
2. Leak testing. Place the manufacturer supplied leak-test accessory (short piece of tubing with a red plug) over the inlet (top port). Press and hold the STOP button and then press the ON button. The display should read "Leak Test, Invert & Push Read." Invert the unit and push Read. Turn the unit upright and allow it to stand. Make sure that the piston is at the top of the cell. Allow the calibrator to stand until the piston falls; this may take as long as 15–20 minutes. The unit will display "Test OK Press Read" if it passes the test. Repeat the leak test with the leak-test accessory over the outlet (bottom port).
3. Calibration. Bios recommends that the unit be recalibrated by the manufacturer annually. The CTC will calibrate the unit as part of their annual service.